

Vanguard's Digital Experience Redesign

A/B Test Results and CX Insights

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Project Date: February 20, 2026

Test Period: March – June 2017



Business Question: Did the Redesign Drive Meaningful Improvement?

Background

Vanguard, redesigned its digital interface with in-context prompts to reduce friction and increase completion. The new experience was tested against the legacy UI across a four-step funnel

Core Question

Did the redesigned UI improve client completion and experience without introducing unwanted friction?

Test Design

Timing: 15 march - 20 June (~ 3 months)

Control Sample Size: 23,532

Test Sample Size: 26,968

Data Overview: Building Our Foundation

Key Data Sources

Our analysis leveraged three distinct datasets to create a comprehensive view of client behaviour and demographics:

- **Client Profiles** (`df_final_demo`): Contains detailed demographic information (age, gender) and account specifics for each client.
- **Digital Footprints** (`df_final_web_data`): A comprehensive record of online client interactions, split into `pt_1` and `pt_2` for merging.
- **Experiment Roster** (`df_final_experiment_clients`): Identifies which clients participated in the A/B test (control vs. test groups).

Data Preparation Steps

To ensure accuracy and create a single source of truth for our analysis, we performed the following key steps:

- **Consolidate Digital Footprint:** Merged `data_pt1` and `data_pt2`, removing duplicate client IDs, visitor IDs, and datetime entries.
- **Filter Experiment Participants:** Ensured only clients explicitly identified as 'control' or 'test' from were included, aligning with the experiment's scope.
- **Integrate All Datasets:** Combined the cleaned digital footprint data with client profiles and experiment rosters, establishing a unified dataset for robust analysis.



Client Profile: Demographically Aligned

48.3

Control Group

Average age (years)

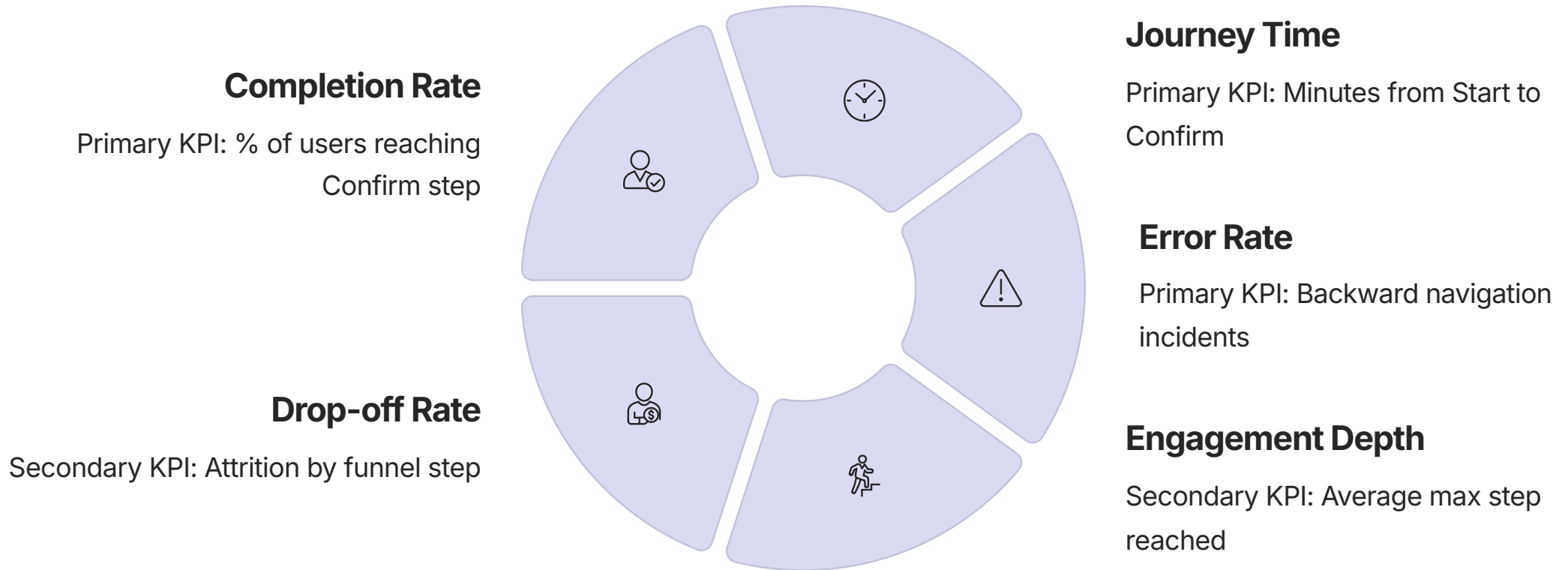
48.8

Test Group

Average age (years)

Welch's t-test detected a statistically significant difference in average age between groups; however, the effect size is negligible and driven by the very large sample. In practice, the age profiles of Control and Test groups are well balanced, meaning the results are unlikely to be influenced by age differences.

KPI Framework: Defining Success



📌 **Success Definition:** Higher completion without increasing friction or errors

Critical Trade-off: Efficiency Costs

Completion Rate

Control: 65.59%

Test: 69.29%

+3.70 pp

The new UI successfully guided more clients through the process

Journey Time Impact

Control: 4.58 minutes

Test: 5.18 minutes

+0.60 min (13.1%)

Test group takes longer to complete, indicating increased friction

Error Rate Spike

Control: 8.85%

Test: 11.64%

-2.79 pp (31.5%)

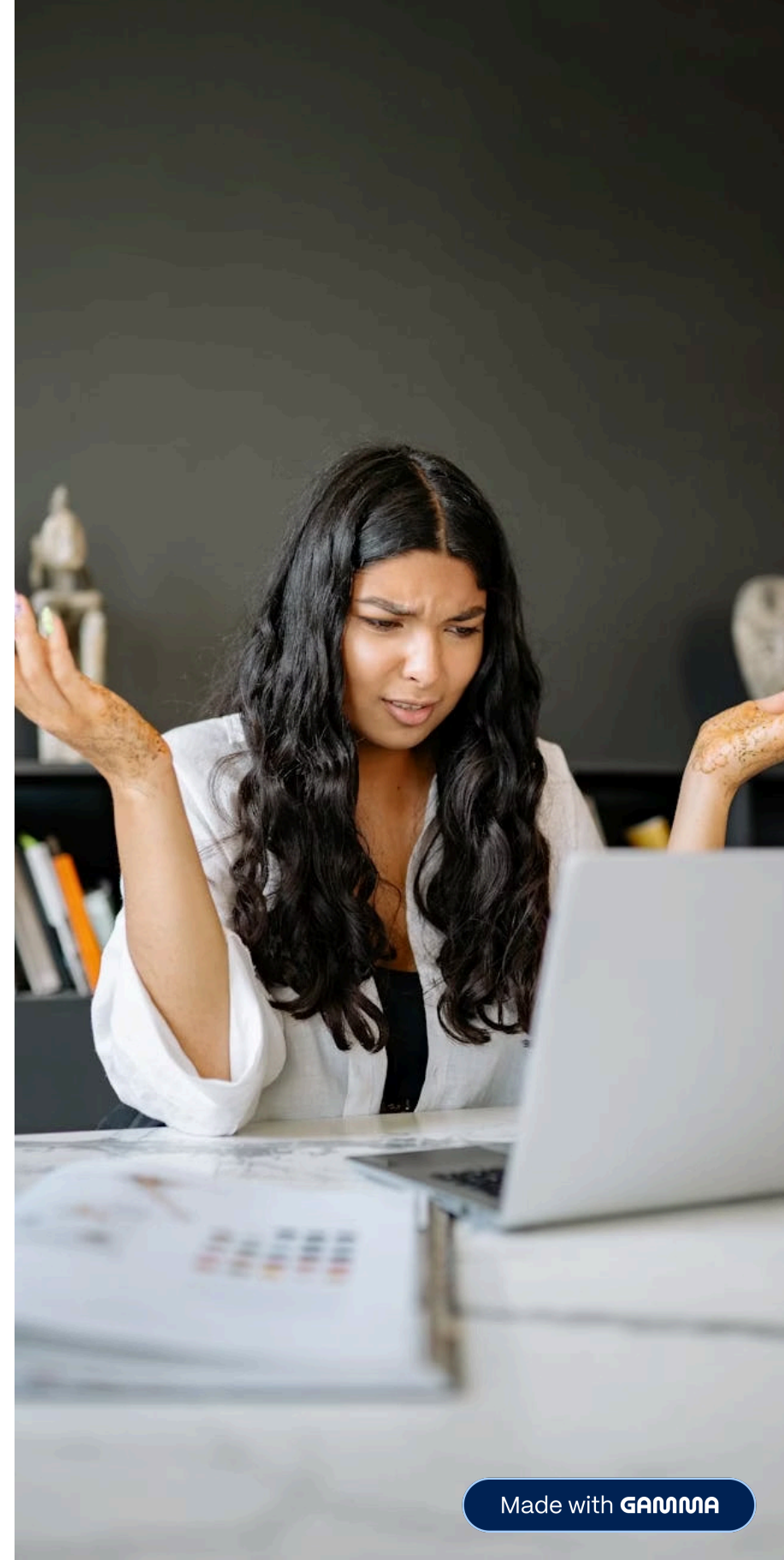
More backward navigation suggests confusion in early steps

Error Hotspot

Highest confusion b/w

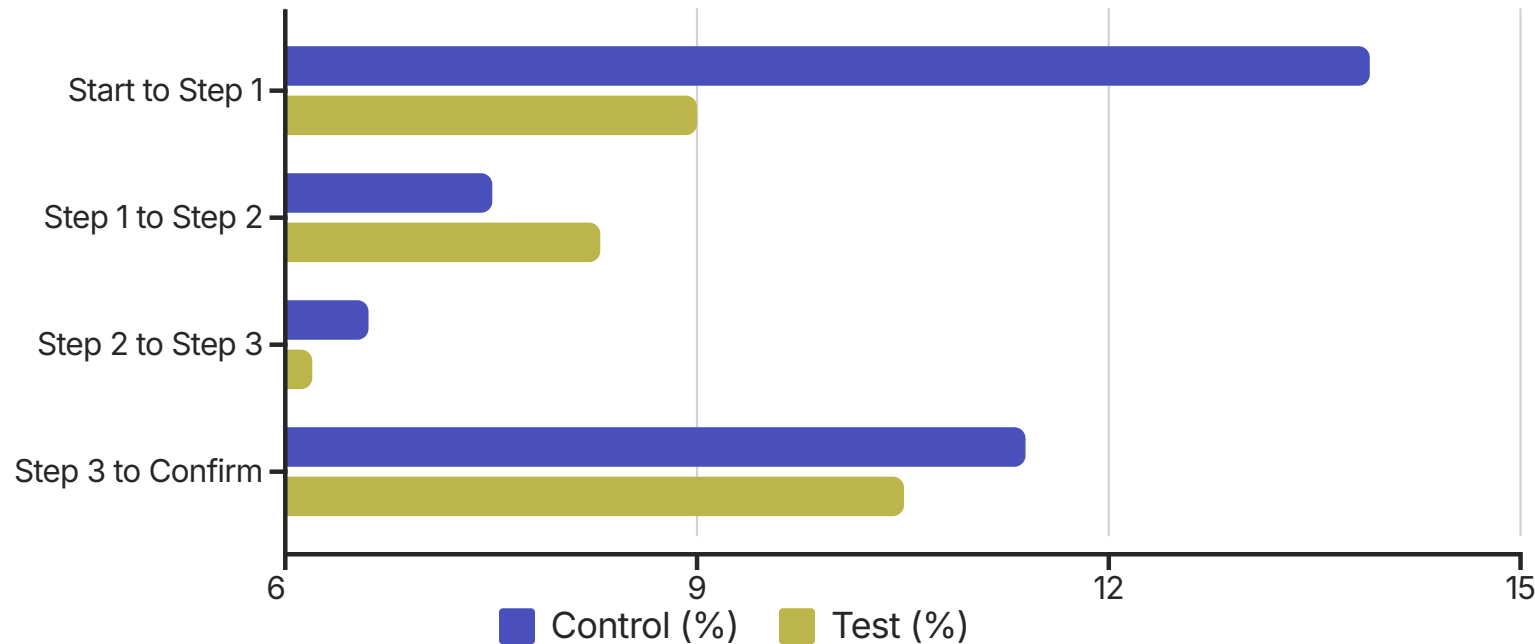
Start → Step 1 in Test experience

In-context prompts may overwhelm users initially



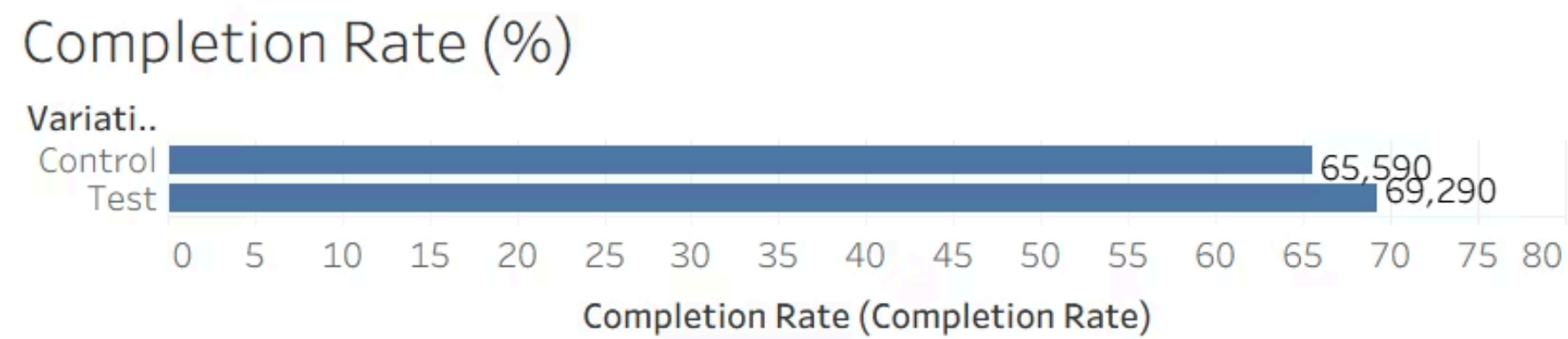
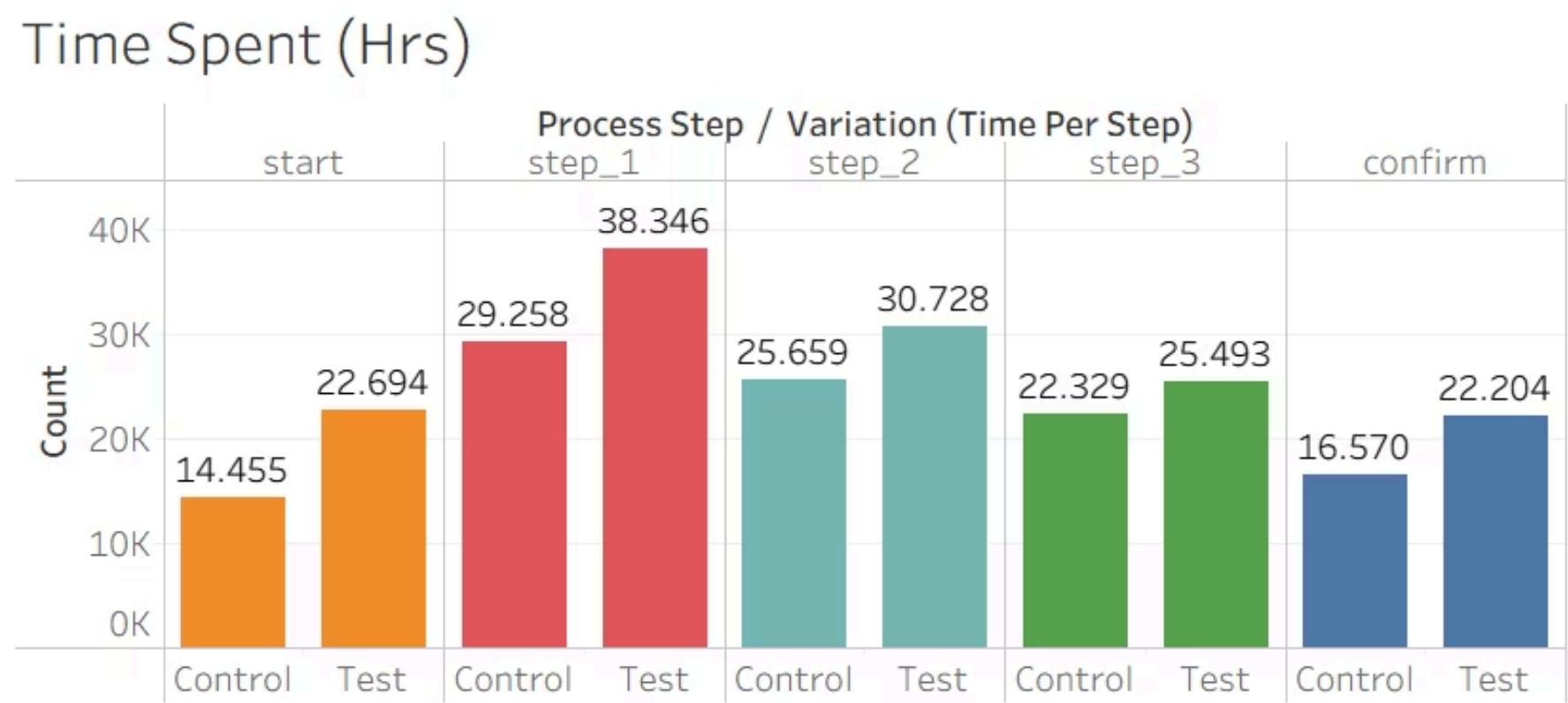
Drop-off Rate Analysis

Understanding user attrition at each stage of the funnel.



- **Initial Engagement:** The test group showed a significantly lower drop-off rate from 'Start' to 'Step 1' (9.0% vs. 13.9%), suggesting the redesigned UI effectively hooks users early.
- **Mid-Funnel Friction:** A slight increase in drop-off for the test group between 'Step 1' and 'Step 2' (8.3% vs. 7.5%) indicates a potential point of friction in the redesigned experience.
- **Improved Completion:** Lower drop-off rates for the test group in the later stages ('Step 2' to 'Step 3' and 'Step 3' to 'Confirm') contribute to the higher overall completion rate, especially at the final hurdle.

Tableau Dashboard (KPIs)



Statistical Validation: Methodology and Findings

Hypothesis Tests Conducted

- Completion rate comparison (z-test/chi-square needed)
- Completion Rate with a Cost-Effectiveness Threshold (5%)
- Age group analysis
- Client tenure analysis (Welch's t-test: $p = 0.558$)

Results Summary

- Completion improvement is meaningful and meets the 5% threshold
- No demographic bias detected in terms of age
- No statistically significant difference in average client tenure



Experiment Effectiveness: Design & Duration

Design Effectiveness

The A/B test was well-structured with a clear control (legacy UI) and test (redesigned UI). Sample sizes were large and comparable, ensuring statistical power:

- Control: 23,532 clients
- Test: 26,968 clients

Demographic analysis confirmed no significant bias, with similar client age profiles (~48.5 years). Results are confidently attributed to UI changes, not user demographics.

Duration Assessment

The test ran for approximately three months (March 15 – June 20, 2017).

This duration was:

- Sufficient for capturing a substantial volume of user behaviour.
- Long enough to mitigate short-term volatility and novelty effects.

While adequate for initial evaluation, it did not capture potential long-term user adaptation or repeat-user learning effects.

Key Takeaway:

Robust experiment, but deeper insights needed

The experiments methodology was sound, sufficiently long, and unbiased. However, richer behavioural and qualitative feedback would enable stronger product decisions going forward.

Conclusions & Recommendations

The experiment:

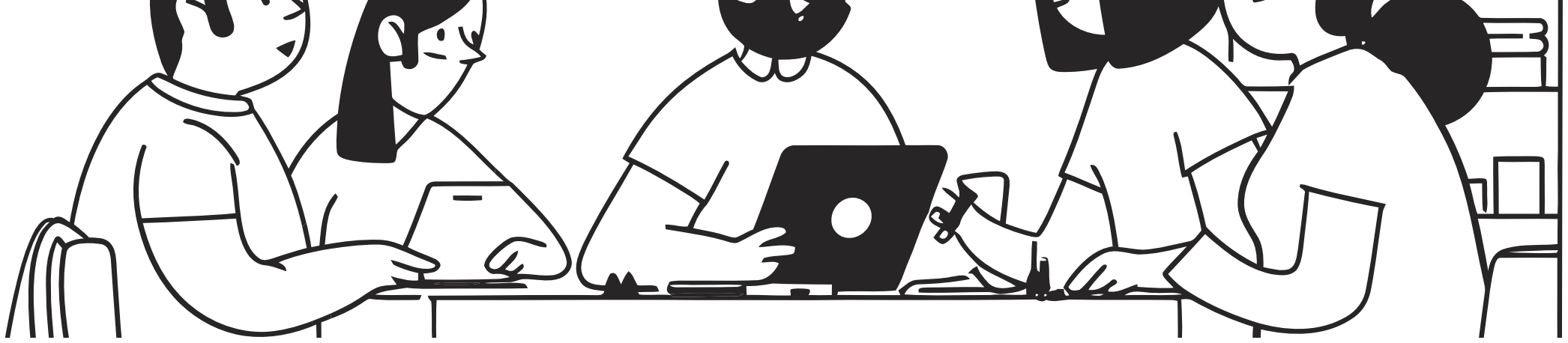
- **is unbiased**; hypothesis testing shows no statistically significant differences between control and test groups.
- A/B test **exceeds the 5% threshold for completion**, which shows that the new UI is justifiable from a cost perspective.
- could benefit from **further iterations** and **complementary qualitative analysis**.

The new User Interface has:

- a **higher completion rate**, guiding a larger number of clients through the process.
- a **longer journey time** and **more backward navigation (error rate)**, indicating increased friction, especially between Steps 1 and 2.

Recommendations:

- Revisit and improve the **user experience between Steps 1 and 2** to decrease drop-off rates.
- **Conduct further rounds** of the A/B test after design updates.
- Collect **deeper qualitative feedback (interviews/surveys/focus groups)** from users for a well-rounded assessment and informed product decisions.



THANK YOU!

Project: Vanguard's Digital Experience Redesign, February 2026

Group B: Andressa, Goldie, Julian, Sarah

Project Kanban Board: <https://trello.com/b/Olwbl9QV/vanguard-projectgroup-b>